

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 636 Deep Learning — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Andrej Karpathy — Neural Networks: Zero to Hero (free)	The primary source for Modules 02, 06, and 07. Builds autodifferentiation and then a transformer from nothing, on video, with every line typed. Do this before anything else.
Zhang, Lipton, Li & Smola — Dive into Deep Learning (free interactive book)	Free, complete, and runnable. Every chapter has working code in several frameworks. The reference for Modules 04, 05, and 08.
Stanford CS231n — Convolutional Neural Networks (free notes and lectures)	The Module 05 reference , and the best free treatment of the practical training material in Modules 03 and 04.
Goodfellow, Bengio & Courville — Deep Learning (free book)	Free in full. The theoretical grounding; dated on architectures and still correct on the fundamentals.
Lilian Weng — Lil'Log (free)	The reference for Module 10. Her posts on diffusion models and on attention are clearer than most papers on the same subjects.
Papers with Code, and the original architecture papers (free)	Read the primary papers for the architectures you use. They are shorter than expected and the ablation tables are where the actual knowledge is.

Module-by-module map

Module 01 — Why Depth

Resource	What it covers for this module
Karpathy — Neural Networks: Zero to Hero, lecture 1 (free)	Start here this week. Builds autodifferentiation from nothing on video; Module 02 assumes you have watched it.
Goodfellow, Bengio & Courville — Deep Learning, chapters 1 and 6 (free)	The §1 framing and the universal approximation discussion, stated carefully rather than as a slogan.
Krizhevsky, Sutskever & Hinton — ImageNet Classification with Deep CNNs (free)	The 2012 result of §3. Read it for the engineering detail — the architectural choices are all justified by ablations.
Battaglia et al. — Relational inductive biases, deep learning, and graph networks (free)	The §2 framing, stated as a general principle. The clearest available treatment of architecture as prior.

Module 02 — Backpropagation

Resource	What it covers for this module
Karpathy — micrograd, and 'The spelled-out intro to neural networks and backpropagation' (free)	The §1 engine, built live on video in two hours. The single best resource for this module.
Baydin, Pearlmutter, Radul & Siskind — Automatic Differentiation in Machine Learning: a Survey (free)	Forward versus reverse mode done properly , and the distinction from symbolic and numerical differentiation.
He, Zhang, Ren & Sun — Deep Residual Learning (free)	The §3 fix. Read the figure showing that a deeper plain network was worse on training error — which is what motivated it.
CS231n — backpropagation and gradient checking notes (free)	The §2 caveats, each explained with the failure it causes.

Module 03 — Optimisation in Practice

Resource	What it covers for this module
Smith — Cyclical Learning Rates and the LR range test (free)	The §1 range test , from the paper that introduced it. Short and immediately applicable.
Loshchilov & Hutter — Decoupled Weight Decay Regularization (free)	The AdamW fix of §2 , with the explanation of what Adam's weight decay was actually doing.
He et al. — Delving Deep into Rectifiers (free)	The §3 initialisation derivation , including the factor of two.
Karpathy — A Recipe for Training Neural Networks (free)	The §4 procedure, in full. The most practically useful single page in this course.

Module 04 — Regularisation and Normalisation

Resource	What it covers for this module
Srivastava et al. — Dropout (free)	The §2 method and the ensemble argument, from the original.
Santurkar, Tsipras, Ilyas & Madry — How Does Batch Normalization Help Optimization? (free)	The §3 correction. The noise-injection experiment is the part to read — it is a model of how to test a mechanistic claim.
Wu & He — Group Normalization (free)	The small-batch problem of §3, measured, with the alternative.
Cubuk et al. — RandAugment (free)	The §4 practice at its current state, with the search space reduced to two hyperparameters.

Module 05 — Convolutional Networks

Resource	What it covers for this module
Stanford CS231n — convolutional networks notes (free)	The reference for §1 and §2 , including the receptive field arithmetic and the architecture history.
He et al. — Deep Residual Learning; Howard et al. — MobileNets (free)	Two of §2's rows, from their papers. The depthwise separable factorisation is worth deriving once.
Liu et al. — A ConvNet for the 2020s (ConvNeXt) (free)	The §3 result. A step-by-step modernisation of a ResNet, with each change measured — which is also a model of how to ablate.
Chaitanya et al. — Interactive Reconstruction of Monte Carlo Image Sequences (free)	The §4 denoiser , and the connection back to CSCE 647.

Module 06 — Sequences and Attention

Resource	What it covers for this module
Karpathy — 'Let's build GPT' (free)	Attention built from scratch on video , with every design choice explained as it is typed. The best resource for §2.
Bahdanau, Cho & Bengio — Neural Machine Translation by Jointly Learning to Align and Translate (free)	Attention's first appearance , bolted onto an RNN. Worth reading to see the problem it was invented to solve.
Christopher Olah — Understanding LSTM Networks (free)	The clearest explanation of §1's gating that exists, with diagrams that actually help.
Dao et al. — FlashAttention (free)	The §3 memory argument — and a direct application of CSCE 735 Module 09's tiling to a problem in this course.

Module 07 — Transformers

Resource	What it covers for this module
Vaswani et al. — Attention Is All You Need (free)	The original. Short, and the ablation table in section 6 is where the design justifications are.
Karpathy — nanoGPT and 'Let's build GPT' (free)	A complete, readable, trainable implementation. Read it alongside §1 — it is about 300 lines.
Hoffmann et al. — Training Compute-Optimal Language Models (Chinchilla) (free)	The §4 correction , with the parameter-to-token ratio derived from experiments.
Schaeffer, Miranda & Koyejo — Are Emergent Abilities of Large Language Models a Mirage? (free)	The metric argument of §4 — a careful demonstration that discontinuous metrics manufacture discontinuous curves.

Module 08 — Training at Scale

Resource	What it covers for this module
Rajbhandari et al. — ZeRO: Memory Optimizations Toward Training Trillion Parameter Models (free)	The §1 memory accounting and the §2 sharding strategy , with the three stages laid out clearly.
Micikevicius et al. — Mixed Precision Training (free)	The §3 loss-scaling argument , including the gradient histograms that show the flush-to-zero problem.
Hugging Face — Performance and Scalability documentation (free)	The practical lever ordering of §1 , with measured effects for each technique.
Chen et al. — Training Deep Nets with Sublinear Memory Cost (free)	Activation checkpointing and the $\sqrt{n}$ result.

Module 09 — Transfer and Representation Learning

Resource	What it covers for this module
Yosinski, Clune, Bengio & Lipson — How transferable are features in deep neural networks? (free)	The §1 measurement — transferability layer by layer, including the negative transfer result.
Radford et al. — Learning Transferable Visual Models From Natural Language Supervision (CLIP) (free)	The §2 objective with the broadest consequences. The zero-shot results section is the surprising part.
Hu et al. — LoRA: Low-Rank Adaptation of Large Language Models (free)	The §3 method, including the intrinsic-rank evidence that motivates it.
Raghu et al. — Transfusion: Understanding Transfer Learning for Medical Imaging (free)	The §4 caution, measured carefully. Required reading before claiming ImageNet pretraining helped.

Module 10 — Generative Models

Resource	What it covers for this module
Lilian Weng — What are Diffusion Models? (free)	The clearest derivation of §3 available , including the connection to score matching.
Kingma & Welling — Auto-Encoding Variational Bayes (free)	The §2 reparameterisation trick , from the paper that introduced it.
Ho, Jain & Abbeel — Denoising Diffusion Probabilistic Models (free)	The §3 formulation that made diffusion practical, with the simplified objective.
Ho & Salimans — Classifier-Free Diffusion Guidance (free)	The §3 guidance mechanism. Two pages of idea and it is behind every text-to-image system.

Module 11 — Neural Rendering

Resource	What it covers for this module
Mildenhall et al. — NeRF: Representing Scenes as Neural Radiance Fields (free)	The §2 method. The positional encoding ablation is the part to read closely.
Kerbl, Kopanas, Leimkühler & Drettakis — 3D Gaussian Splatting for Real-Time Radiance Field Rendering (free)	The §3 method , with the differentiable rasteriser described in enough detail to implement.
Müller, Evans, Schied & Keller — Instant Neural Graphics Primitives (free)	The §2 representation change — hash grids, and the seconds-not-days result.
Tewari et al. — Advances in Neural Rendering (free survey)	The landscape, including the §4 limitations stated honestly by people working in it.

Module 12 — What Networks Fail At

Resource	What it covers for this module
Ilyas et al. — Adversarial Examples Are Not Bugs, They Are Features (free)	The §1 reframing , with the dataset experiment. One of the more genuinely surprising results in the field.
Geirhos et al. — Shortcut Learning in Deep Neural Networks (free)	The §2 survey — the documented cases, the mechanism, and what to do about it.
Adebayo et al. — Sanity Checks for Saliency Maps (free)	The §4 result. The weight-randomisation test, and which methods fail it.
Hendrycks & Dietterich — Benchmarking Neural Network Robustness to Common Corruptions (free)	The §4 stress-test benchmark, and evidence that it predicts deployment behaviour.

Module 13 — Shipping a Model

Resource	What it covers for this module
Hinton, Vinyals & Dean — Distilling the Knowledge in a Neural Network (free)	The §2 'dark knowledge' argument , from the paper that named it.
Jacob et al. — Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference (free)	The §2 quantisation scheme, including quantisation-aware training.
Kwon et al. — Efficient Memory Management for Large Language Model Serving (PagedAttention / vLLM) (free)	The §3 KV cache problem , and a solution borrowed directly from virtual memory (CSCE 611).
Blalock et al. — What is the State of Neural Network Pruning? (free)	The §2 caution — a survey finding that pruning results are frequently not comparable and that reported speedups are often theoretical.